

Standardizing acute toxicity data for use in ecotoxicology models: influence of test type, life stage, and concentration reporting

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Abstract Ecotoxicological models generally have large data requirements and are frequently based on existing information from diverse sources. Standardizing data for toxicological models may be necessary to reduce extraneous variation and to ensure models reflect intrinsic relationships. However, the extent to which data standardization is necessary remains unclear, particularly when data transformations are used in model development. An extensive acute toxicity database was compiled for aquatic species to comprehensively assess the variation associated with acute toxicity test type (e.g., flow-through, static), reporting concentrations as nominal or measured, and organism life stage. Three approaches were used to assess the influence of these factors on log-transformed acute toxicity: toxicity ratios, log-linear models of factor groups, and comparison of interspecies correlation estimation (ICE) models developed using either standardized test types or reported concentration type. In general, median ratios were generally less than 2.0, the slopes of log-linear models were approximately one for well-represented comparisons, and ICE models developed using data from standardized test types or reported concentrations did not differ substantially. These results indicate that standardizing test data by acute test type, reported concentration type, or life stage may not be critical for developing ecotoxicological models using large datasets of log-transformed values.

Keywords Ecotoxicological models · Data standardization · Acute toxicity · Life stage · Acute test type · Measured concentrations

Introduction

A new paradigm of toxicology emphasizes the transition from animal testing to computational approaches (NAS 2007). The emphasis on reduced toxicity testing is occurring simultaneously with increasing demands for more comprehensive understanding of toxicity pathways and reduced uncertainty in risk assessment (Bodar et al. 2002; Bradury et al. 2004). To address increasing data needs concomitant with reduced or infeasible animal testing, ecotoxicological models have been developed to allow estimation of acute toxicity in untested species (Raimondo et al. 2007a) and for deriving hazard levels based on species sensitivity relationships (Dyer et al. 2008). Despite their utility, ecotoxicological models generally have large data requirements and are frequently based on existing information from diverse sources (e.g., ECOTOX; <http://www.epa.gov/ecotox/>) and testing conditions. High quality assurance and data standardization will be critical to ensure that toxicological models developed from compiled datasets produce consistent and reliable predictions.

Datasets compiled for ecotoxicological models may include data that vary in test conditions, use of measured or nominal concentrations, organism life stage or size, and chemical formulation, potentially introducing large amounts of variability into the models. Standardizing data for toxicological models will reduce extraneous sources of variation so models more accurately reflect intrinsic relationships. Datasets maybe standardized, e.g., by including

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only juvenile data to control the variation associated with organism life stage. Guidance has been suggested for standardizing acute toxicity data for different taxa (Buckler et al. 2003), and used for several modeling efforts (Asfaw et al. 2003; Raimondo et al. 2007a, 2008). However, for toxicological models that use log-transformed toxicity data the influence of some factors may be relatively minimal. For example, Mayer and Ellersieck (1986) determined that the mean ratio of toxicity values derived from static to flow-through tests was 2.0, which suggested an insignificant amount of variation due to test type. Determining factors with the largest influence on toxicity test results is important, because limiting datasets to control for factors that contribute minimal variation may impact model robustness by reducing the amount of data available for model development.

In this study, an extensive acute toxicity database was compiled for aquatic species to assess the magnitude of the variation associated with three factors that have been known to influence reported toxicity values: organism life stage, acute test type (e.g., static, flow-through), and reporting concentrations as measured or nominal. Organism life stage and acute test type (e.g., static, flow-through) can be an important source of variation for specific taxa and chemicals (Mayer and Ellersieck 1986; Gerhardt 1992; Fisher et al. 1994; Gaikowski et al. 1996; Kim et al. 2008), whereas the influence of concentration reporting has not been quantitatively investigated. The variability associated with organism life stage, acute test type (e.g., static, flow-through), and measured or nominal concentrations was quantified to determine if these three factors significantly influenced log-transformed toxicity values within an extensive and diverse dataset of chemicals and species. Log transformed values were assessed because they are frequently used in model development (Dyer et al. 2006, 2008; Raimondo et al. 2007a, b; Wang et al. 2008) and the influence of some factors may be substantially reduced through data transformations. Interspecies correlation estimation (ICE) models were developed to compare models with identical species and chemical composition that varied only in acute test type or reported concentration (measured or nominal). ICE models are log-linear least square regression models that describe the relationship of acute toxicity for a range of chemicals tested in two species (Raimondo et al. 2007a, b). ICE models were selected because they are developed from database compilations with relatively small data requirements that allowed for the comparison of models from highly standardized data. Factors that were species specific (e.g., pH, temperature) were not addressed, but have been shown elsewhere to generally have minimal variation (Mayer and Ellersieck 1986). This research is intended to benefit future model

development efforts by determining standardization needs and criteria for diverse datasets.

Materials and methods

Database development

We compiled a database of 11,943 acute toxicity records that included 432 species and 1,600 chemicals (Table 1). Data sources were the United States Environmental Protection Agency (USEPA) Office of Pesticide Programs ecotoxicity database, ECOTOX/ACQUIRE (<http://cfpub.epa.gov/ecotox>; accessed May 2007), Ambient Water Quality Criteria documents (USEPA 1986, 1996a), and the US EPA Gulf Ecology Division ICE database (Raimondo et al. 2007a, 2008). The database only included chemicals with 90% or greater active ingredient and acute toxicity test conditions that met requirements for each species (ASTM 1980, 2007). Data included 96 h median lethal concentration (LC50) or median effect concentration (EC50) values for fish and most invertebrates (mysids, mussels), and 48 h toxicity values for daphnids and midges. For freshwater species, toxicity values were adjusted for metals to a hardness of 50 mg/l, pentachlorophenol to pH 6.8, and ammonia to pH 8.0 (USEPA 2002). Duplicate records collected from multiple sources and open-ended toxicity values ($>100 \mu\text{g/l}$ or $<100 \mu\text{g/l}$) were excluded from the database.

The life stage of each species was broadly defined as larvae, juvenile, or adult for all toxicity records; the egg life stage was excluded from the data compilation because of its relatively low occurrence in the database. The specified life stage was used when it was reported in the original source. When a specific stage was not identified in the original source, as occurred for 44% of records, natural history and organism length were used to determine life stage (Supplemental Information 1). Only the larval stage of amphibians and aquatic insects were included in the database. Mollusk glochidia were categorized as larva and oyster spat were classified as juvenile. Fish larvae included hatchlings through full fin development. Juvenile fish were those with full fin development lacking sexual maturity and adult fish were those that were sexually mature. In cases where only a weight was provided for a fish species, life stage was determined using length-weight regressions found in Fish Base life history tool (Froese and Pauly 2008). When length-weight regressions were not available or adequate information was not provided, age class was designated as unknown and these records were not included in the database.

Table 1 Number of datapoints, number of species, and number of chemicals in dataset and for measured or nominal, acute test type, and life stage data subsets

	Taxa	Overall	Measured	Nominal	Static	Static renewal	Flow-through	Larvae	Juveniles	Adults
Number of records	All taxa	11,943	3,471	8,472	7,765	814	3,364	1,023	9,519	1,401
	Amphibia	124	45	79	64	36	24	0	124	0
	Annelid	83	23	60	62	13	8	6	34	43
	Crustacea	2,708	639	2,069	2,003	274	431	73	2,003	632
	Fishes	8,031	2,513	5,518	4,928	409	2,694	869	6,564	598
	Insecta	511	84	427	469	5	37	0	511	0
	Mollusca	486	167	319	239	77	170	75	283	128
Number of species	All taxa	432	227	327	309	147	163	106	301	171
	Amphibia	22	9	18	13	15	6	0	22	0
	Annelid	15	7	12	10	6	4	3	5	13
	Crustacea	106	57	80	71	43	37	19	53	74
	Fishes	165	97	125	119	52	85	63	133	52
	Insecta	56	23	43	46	4	19	0	56	0
	Mollusca	68	34	49	50	27	12	21	32	32
Number of chemicals	All taxa	1,600	777	1,098	997	253	911	202	1,526	388
	Amphibia	55	21	41	31	18	14	0	55	0
	Annelid	35	14	26	31	6	4	6	21	20
	Crustacea	839	200	738	672	143	222	31	768	206
	Fishes	1,316	715	812	703	139	823	185	1,242	233
	Insecta	164	47	134	153	4	23	0	164	0
	Mollusca	196	37	170	52	29	136	28	151	51

Data comparisons

To compare toxicity test results reported as either measured or nominal concentrations, the data were subset into measured and nominal concentrations paired by species and chemical (227 paired observations). The subset contained only juvenile life stages to minimize variation associated with organism life stage. If there were multiple toxicity values for a specific chemical, species, and reported concentration type, the geometric mean was used as the acute toxicity value. The median and range of nominal to measured ratios were calculated to provide an estimate of variability of untransformed data. A log-linear model was used to compare toxicity reported as nominal and measured concentrations. Toxicity values reported as nominal concentrations were the independent variable and those reported as measured concentrations were the dependent variable to determine if nominal concentrations could provide adequate surrogacy for measured concentrations. The slope of the log-linear model was compared to 1.0 using a *t*-test to determine if log-transformed measured and nominal concentrations exhibited a one to one relationship (Sokal and Rohlf 1998). This analysis was first performed using only flow-through tests and then performed using data from all acute test types. Analyses were conducted for all taxa combined and for fish, crustaceans, and insects

using data from all acute test types and fish and crustaceans using flow-through tests only.

To compare acute test types, three data subsets were created that included toxicity data paired by chemical and species from static and static-renewal tests, static and flow-through tests, and static-renewal and flow-through tests. Subsets contained only juvenile data to minimize variation associated with organism life stage. If there were multiple toxicity values for a specific chemical, species, and acute test type, the geometric mean was used as the acute toxicity value. Ratios were calculated as static to flow-through, static to static-renewal, and static-renewal to flow-through, with the numerator being the test type expected to yield a higher toxicity value. Data from each pair of acute test types were fit to a log-linear model and slopes were compared to 1.0. Flow-through was the dependent variable in the two models in which it was included to determine if the other test types could provide adequate surrogacy for this test type. In the comparison of static and static-renewal tests, static-renewal was the dependent variable. This analysis was performed for all taxa combined, and where data were available, for fish, crustaceans, and insects. Analyses were first conducted using measured concentrations only and then with both measured and nominal concentrations.

To compare life stages, three data subsets were created that paired life stages by common chemical and species and

were composed of larvae and juveniles, larvae and adults, and juveniles and adults. Test data included both measured and nominal concentrations from all acute test types because there were insufficient data to create more highly standardized datasets. If there were multiple toxicity values for a specific chemical, species, and life stage, the geometric mean was used as the acute toxicity value. Ratios were calculated as the older life stage divided by the younger life stage to yield potentially high ratios associated with greater sensitivity of earlier life stages (McKim 1985). Data for paired life stages were fit to log-linear models as in the above analyses for all taxa combined, and for annelids, crustaceans, fish, and mollusks when data were available. In these models, the juvenile life stage was the independent variable to determine if the toxicity data for juveniles, which are the most abundant test data, was representative of larval or adult life stages. In the model of larvae and adults, the adult life stage was the independent variable.

ICE model comparisons

ICE models are log-linear least square regression models that describe the relationship between the acute toxicity of a range of chemicals tested in two species (Raimondo et al. 2007a). Each ICE model is developed by pairing toxicity data by common chemical for two species: the surrogate and the predicted. The relationship of species sensitivity described by the model is used to estimate the acute toxicity to the predicted taxa from the known toxicity of the surrogate species. A diversity of ICE models have been developed and validated for extrapolation of toxicity values for aquatic organisms and wildlife (Raimondo et al. 2007a, b).

The influence of using test data reported as only nominal concentrations was determined by comparing ICE models developed from either datasets of measured or nominal concentrations. For each surrogate and predicted species pair, two ICE models with identical chemical composition were developed. One model contained only data reported as nominal and the second contained only data reported as measured concentrations. Models contained juvenile data only. There were insufficient data to further standardize these datasets by acute test type, so data from all test types were combined. For each surrogate and predicted species pair, the slopes from ICE models developed from measured and nominal concentrations were compared using a *t*-test, and whole model comparisons were conducted by Analysis of Covariance (Sokal and Rohlf 1998). Only significant ($P < 0.05$) ICE models were included in the analyses.

The validity of using toxicity reported as either nominal or measured concentrations in ICE models was further examined by using the toxicity reported from measured concentrations as input for models developed from nominal

concentrations, and vice versa, for each surrogate-predicted species pair. For example, for an ICE model developed from nominal data with a bluegill surrogate and fathead minnow predicted species, bluegill toxicity reported from measured concentrations was used as independent model input to predict the fathead minnow toxicity. The toxicity values predicted from the ICE models developed from nominal (or measured) concentrations were compared to the respective toxicity values reported from measured (or nominal) concentrations. Model accuracy was determined as the percentage of datapoints predicted within 5-fold of the reported value, selected based on the range of inter-laboratory variation estimated for acute toxicity (Fairbrother 2008).

Two sets of ICE models were developed from data reported from either static or flow-through tests to determine the influence of test type. Models developed for each surrogate and predicted species pair had identical chemical composition. Only data for juveniles were used in the models, however, there were insufficient data to further standardize these data sets by measured concentrations or to develop ICE models from static-renewal tests. Models developed from static and flow-through tests were compared for each surrogate and predicted species pair as above. The ability of ICE models developed from static tests to predict toxicity reported from flow-through tests, and vice versa, was verified using the methods described above. There was insufficient data to develop unique models for different life stages to determine its influence on ICE models.

Results

The dataset allowed the analyses of acute test type and reported concentration type to be highly standardized, using data for juveniles only and flow-through tests or measured concentrations to isolate the effects of reported concentration type or test type, respectively. The analyses of reported concentration and acute test type were conducted prior to life stage analysis to determine if it was necessary to standardize by these factors when comparing life stages. Overall, these analyses indicated that reported concentration type and acute test type had little effect on the variability of log-transformed toxicity and all data from all test and concentration types were included in the life stage analysis. Additionally, standardizing the life stage datasets by both acute test type and reported concentration type resulted in sample sizes that were too small to comprehensively assess the variability associated with taxa-specific life stages.

Comparisons of measured and nominal concentrations indicated that reported test concentration type had little

Table 2 log-linear models developed from paired measured and nominal concentrations using flow-through tests only and all acute test types

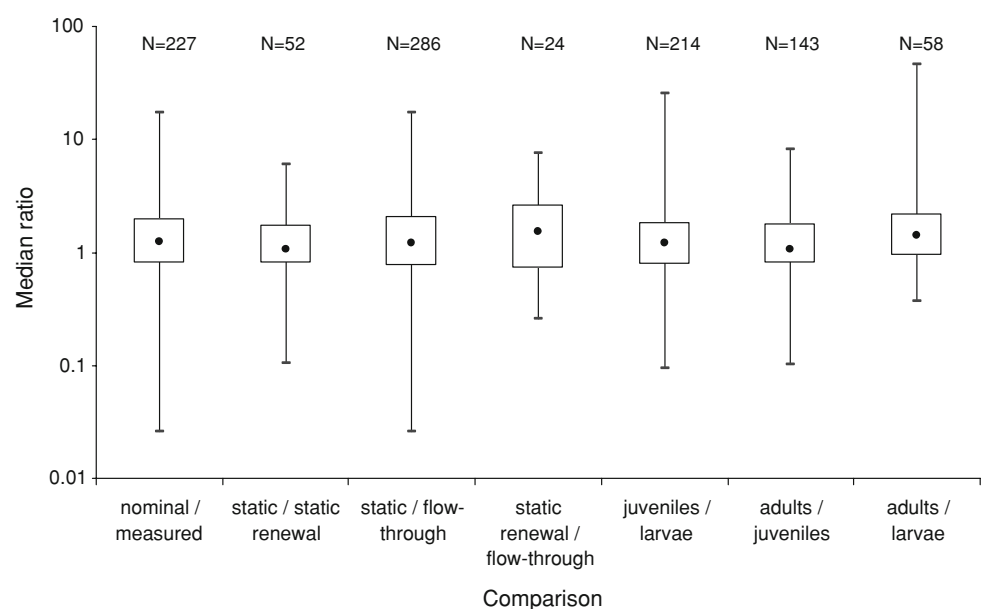
Taxa	<i>N</i>	Intercept	Slope	SE slope	<i>P</i>	<i>R</i> ²	<i>t</i>
Flow-through test data only							
all	27	0.06	0.99	0.030	<0.001	0.98	−0.26
Fishes	23	0.09	0.98	0.036	<0.001	0.97	−0.52
Crustacea	4	0.03	1.14	0.012	<0.001	1.00	11.84 ^a
All acute test types							
all	227	0.02	0.97	0.014	<0.001	0.96	−2.36 ^a
Fishes	163	−0.07	1.00	0.016	<0.001	0.96	0.14
Crustacea	56	0.09	0.90	0.028	<0.001	0.95	−3.61 ^a
Insecta	6	0.38	0.85	0.153	0.005	0.89	−0.98

^a Significant at $P < 0.05$

effect on the variability of log-transformed toxicity. Using the highly standardized dataset of juveniles and flow-through acute tests only, there were 27 paired tests for comparing log-transformed measured and nominal concentrations. The median nominal to measured ratios ranged from 0.92 to 0.99 for fish, crustaceans, and all taxa combined. Minimum and maximum ratios for fish, crustaceans, and all taxa combined were 0.32 and 4.13, respectively. The slopes of the log-linear models ranged from 0.98 (fish, $N = 23$) to 1.14 (crustaceans, $N = 4$) and were not significantly different than 1.0 for all taxa combined and fish, but were different from 1.0 for crustaceans (Table 2). Using only juvenile data and including all acute test types there was a total of 227 paired measured and nominal toxicity records for all taxa. The median nominal to measured ratios ranged from 0.9 to 1.5 for individual

taxonomic groups and all taxa combined, with the median of all taxa combined at 1.25 (Fig. 1). Minimum and maximum ratios for individual taxonomic groups and all taxa combined were 0.03 and 17.15, respectively. The interquartile ranges of these comparisons were bound by 0.64 and 2.53. The slopes of the log-linear models ranged from 0.85 (insects, $N = 6$) to 1.00 (fish, $N = 163$) and were significantly different from 1.0 for all taxa combined (0.97) and crustaceans (0.90), but not for fish and insects (Table 2; Fig. 2a). There were 12 surrogate and predicted species pairs for which ICE models could be developed using only measured or nominal concentrations. Models contained between five and 12 toxicity records. There were no ICE models developed from toxicity data reported as measured concentrations that were significantly different from the respective model developed from toxicity data reported as nominal concentrations (Supplemental Information 2). ICE models developed from toxicity reported as either nominal or measured concentrations predicted the toxicity of the alternative concentration type within 5- and 10-fold of the actual value for 82 and 91%, respectively, of all values ($N = 192$). Nominal to measured ratios demonstrate minimal variation in untransformed toxicity data was attributed to reported concentration type for most records. Based on the log-linear relationship of nominal and measured concentrations and ICE model comparisons, reported concentration type appears to have little influence on the variability of log-transformed toxicity values.

Acute test type also had little effect on the variability of log-transformed toxicity data. Using only juvenile life stages and test data reported as measured concentrations, the median ratios were less than 2.0 for comparisons of test types across all taxa. The range of ratios for all test types

Fig. 1 Median ratios of comparisons including all taxa and test types. The whiskers are the range and boxes are the interquartile range of ratios for each group

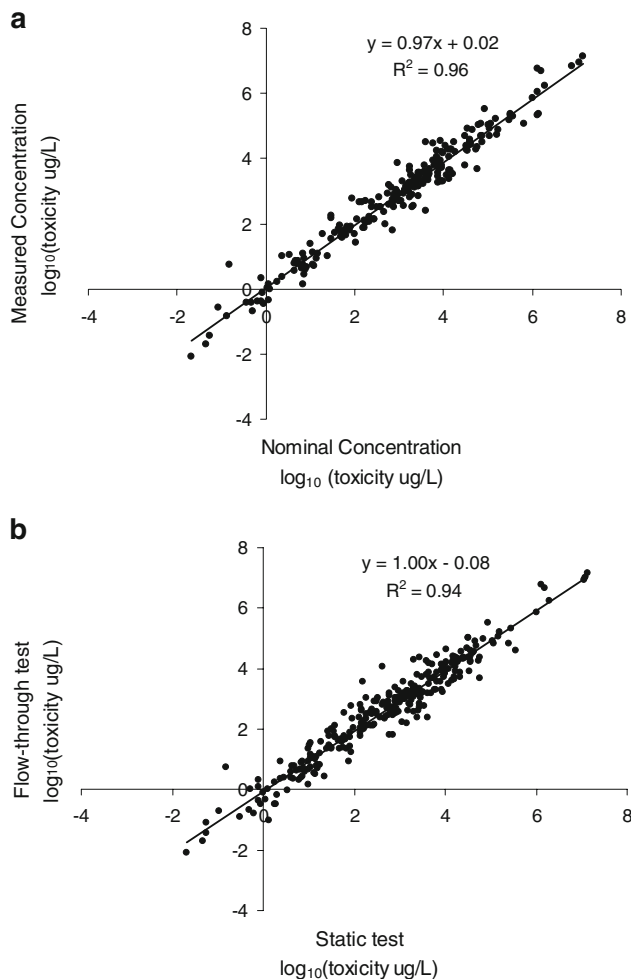


Fig. 2 Linear models comparing log-transformed **a** measured and nominal concentrations using all acute test types and all taxa and **b** static and flow through tests using toxicities reported as both measured and nominal concentrations and all taxa

across all taxa was 0.1–9.1. The slopes of the log-linear models of two test types ranged from 0.95 (crustaceans, static renewal vs. flow-through, $N = 4$) to 1.14 (Crustaceans, static vs. static renewal, $N = 6$), and no slopes were significantly different from one (Table 3). With the exception of the comparison of static and flow-through for all taxa ($N = 49$) and fish ($N = 38$), using measured data only resulted in small sample sizes ($N < 11$). Using data reported for only juvenile life stages and both measured and nominal concentrations, the median ratios of all test type comparisons were less than 2.1 for individual taxa and all taxa combined (Fig. 1). The range of ratios for all test types across all taxa was 0.03–17.15 with an overall interquartile range bound by 0.29 and 4.92. Slopes of the log-linear models ranged from 0.86 (fish, static vs. static renewal, $N = 21$) to 1.01 (fish, static vs. flow-through, $N = 235$) and were not different from one for any test type comparison (Table 3; Fig. 2b). ICE models were

developed using either static or flow-through test types for 24 surrogate-predicted species pairs and contained between 3 and 20 toxicity records. ICE models developed from static toxicity test data were not significantly different from the respective model developed from flow-through toxicity test data (Fig. 3; Supporting Information 2). ICE models developed from toxicity reported from flow-through or static tests predicted the toxicity of the alternative test type within 5- and 10-fold of the actual value for 85 and 93%, respectively, of all values ($N = 376$). Based on log-linear models of the acute test types and the ICE model comparisons, acute test type appears to have little influence on the variability of log-transformed toxicity values.

The influence of life stage was variable across taxonomic groups, but generally contributed minimal variation to log-transformed toxicity values. The median toxicity ratios were less than 2.1 for all taxonomic groups and life stages, with fish containing the greatest variability (juvenile to larvae ratios range = 0.03–25.31, juvenile to adult ratio range = 0.10–7.62, adult to larvae ratio range = 0.37–45.66). Ratios of other taxonomic groups ranged from 0.15 to 8.26. Interquartile ranges of all taxa ratios were bound by 0.23 and 3.41. Median ratios for the different life stages were all greater than 1 (Fig. 1), indicating that larvae were generally more sensitive than juveniles and adults, and juveniles were more sensitive than adults for all taxa except mollusks (data not shown). The slopes of log-linear models developed from data for different life stages ranged from 0.53 (crustaceans, juveniles vs. larvae, $N = 4$) to 1.50 (Annelids, adults vs. larvae, $N = 4$). The life stage comparisons that contained 30 or more datapoints had slopes near 1.0 and ranged from 0.96 (all taxa, juveniles vs. adults, $N = 143$) to 1.04 (fish, adults vs. larvae, $N = 43$). Of these, only the slope for juveniles and adults of all taxa (0.96) was significantly different from 1.0 (Table 4).

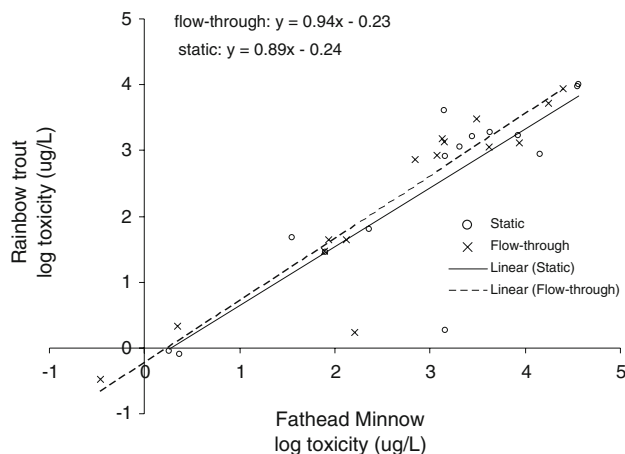
Discussion

Determinants of variability in toxicity test data have been of interest for over 30 years in aquatic toxicology (Sprague 1973; Hedtke et al. 1986; Mayer et al. 1994; Raimondo et al. 2007c). Numerous factors have been considered important determinants of within-species variability, including chemical formulation, test conditions, and organism size or life stage. Mayer and Ellersieck (1986) assessed the effect of test conditions (temperature, pH, hardness), acute test type and size of test animals on acute toxicity of aquatic organisms and reported that organism life stage and acute test type were among the largest sources of variation. Testing conditions such as temperature and hardness are known to be important sources of toxicity test variation (Mayer and Ellersieck 1986).

Table 3 Log-linear models developed from paired acute test types using measured concentrations only and both measured and nominal concentrations

Comparison	Taxa	N	Intercept	Slope	Standard error of slope	P	R ²	t
Measured concentrations only								
Static versus static renewal	All	11	−0.34	1.08	0.100	<0.001	0.93	0.84
	Crustacea	6	−0.64	1.14	0.053	<0.001	0.99	2.62
Static versus flow-through	All	49	−0.08	1.00	0.039	<0.001	0.93	0.06
	Fishes	38	−0.05	1.00	0.043	<0.001	0.94	−0.10
	Crustacea	9	−0.14	0.99	0.113	<0.001	0.92	−0.05
Static renewal versus flow through	All	6	0.02	0.95	0.121	0.001	0.94	−0.38
	Crustacea	4	0.09	0.95	0.204	0.043	0.91	−0.26
Measured and nominal concentrations								
Static versus static renewal	All	52	0.13	0.95	0.038	<0.001	0.93	−1.21
	Fishes	21	0.62	0.86	0.072	<0.001	0.88	−1.95
	Crustacea	26	0.06	0.92	0.044	<0.001	0.95	−1.73
Static versus flow-through	All	286	−0.08	1.00	0.015	<0.001	0.94	−0.07
	Fishes	235	−0.08	1.01	0.015	<0.001	0.95	0.43
	Crustacea	44	−0.11	0.94	0.052	<0.001	0.89	−1.13
	Insecta	6	0.39	0.92	0.380	0.072	0.60	−0.20
Static renewal versus flow through	All	24	−0.06	0.96	0.052	<0.001	0.94	−0.69
	Fishes	15	−0.17	0.97	0.068	<0.001	0.94	−0.49
	Crustacea	8	0.02	1.00	0.113	<0.001	0.93	0.01

No comparisons were significant at the $P < 0.05$ level

**Fig. 3** Example ICE models developed using static and flow-through acute test data for fathead minnow and rainbow trout

However, they are often confounded with species-specific or chemical-specific test requirements and may have a limited influence on toxicity test results if test conditions are appropriate for the species and chemical. This study focused on the intraspecific factors that were determined to be significant sources of variability by Mayer and Ellersieck (1986) as well as the influence of reported test concentration, which has not been previously analyzed. While we recognize these factors contribute significant sources of variation in untransformed data applied in chemical or

species specific risk assessments, our objective was to address the challenge of standardizing data for developing models with large data requirements compiled from diverse data sources. We used a high quality dataset of almost 12,000 acute toxicity values to determine if these intra-specific factors should be standardized prior to use in ecotoxicological models which are based on transformed data.

Three approaches were used to assess the influence of acute toxicity test type and reporting toxicity values as either measured or nominal concentrations: toxicity ratios, log-linear models comparing the three factors, and comparison of ICE models developed using specific test types or reported concentrations. The influence of organism life stage was assessed using toxicity ratios and log-linear models comparing larval, juvenile, and adult life stages. The use of toxicity ratios is a classical approach that has been used to compare acute test type (Mayer and Ellersieck 1986) and life stage (Hutchinson et al. 1998). Over all the comparisons, the median ratios were generally less than 2.0 for all taxa. Despite showing limited variability, toxicity ratios are used here to provide an assessment of variation of untransformed data and may not be the most relevant basis for determining if a factor is important for models of log-transformed data. This is more effectively demonstrated by the log-linear models comparing two factors. Log-linear models were used in this study to determine if log-

Table 4 Log-linear models developed from paired life stages

Comparison	Taxa	<i>N</i>	Intercept	Slope	Standard error of slope	<i>P</i>	<i>R</i> ²	<i>t</i>
Juveniles versus larvae	All	214	−0.03	0.99	0.015	<0.001	0.95	−0.88
	Crustacea	4	0.29	0.53	0.034	0.004	0.99	−13.96 ^a
	Fishes	202	−0.04	0.99	0.016	<0.001	0.95	−0.54
	Mollusca	8	0.18	0.92	0.107	<0.001	0.92	−0.71
Juvenile versus adults	All	143	0.15	0.96	0.015	<0.001	0.97	−2.67 ^a
	Annelid	6	0.53	0.86	0.055	<0.001	0.98	−2.45
	Crustacea	30	0.16	1.02	0.033	<0.001	0.97	0.71
	Fishes	98	0.10	0.97	0.018	<0.001	0.97	−1.61
	Mollusca	9	0.10	0.94	0.096	<0.001	0.93	−0.67
Adults versus larvae	All	58	−0.18	0.99	0.029	<0.001	0.95	−0.21
	Annelid	4	−1.77	1.50	0.594	0.128	0.76	0.84
	Crustacea	9	0.06	0.85	0.039	<0.001	0.99	−4.00 ^a
	Fishes	43	−0.35	1.04	0.038	<0.001	0.95	0.96

^a Significant at $P < 0.05$

transformed values obtained from different acute test types, life stages, or reported concentrations were equivalent. ICE models were used as example ecotoxicological models that are developed from diverse data sources and may contain toxicity data that vary in these factors. The slope of log-linear models of acute test type, life stage, and measured and nominal concentrations were approximately one for well-represented comparisons. Additionally, standardizing datasets by acute test type or reported concentration type did not result in ICE models that differed substantially. These results indicate that standardizing test data by acute test type, reported concentration type, or life stage may not be critical for developing ecotoxicological models using large datasets of log-transformed values.

Most current guidelines for performing acute toxicity tests recommend renewal of test solutions to maintain water quality and exposure concentrations (ASTM 2007), but the majority of historic data have been collected in static acute toxicity tests (e.g., Mayer and Ellersieck 1986; USEPA 1996a). For example, a broad query of over 200,000 acute toxicity records for all aquatic taxa in the Ecotox database (USEPA 2007) showed that 62% were static tests. Static test data are invaluable because they contain a large diversity of chemicals and species, and thus databases compiled for model development are likely to include a large contribution from static tests. Static tests comprised 65% of the acute toxicity records used in the current study. This study compared static and flow-through tests for 286 chemical and species pairs that included a wide range of chemicals and species. We found median static to flow-through and static to renewal ratios were approximately 2.0, which is consistent with comparison of acute test types in previous studies (Mayer and Ellersieck

1986; Gerhardt 1992; Moles 1998). Log-linear models of acute test types showed that well-represented samples ($n > 30$) had slopes that were not significantly different than one, and indicated that log-transformed toxicity data acquired from different test types maintain an approximate 1–1 relationship. This was further demonstrated by ICE models developed from static and flow-through test data. Models were identical in chemical composition for each surrogate and predicted species pair and varied only in acute test type. While there was variation among ICE models developed for static and flow-through tests, the models were not found to be significantly different from each other and models developed from one test type predicted the reported toxicity of the other test type with high certainty. Overall, these results indicate that toxicity values determined in all three acute test types maybe included in ecotoxicological models.

The majority of current test guidelines recommend reporting toxicity results determined from measured concentrations of the test chemical, rather than basing toxicity values on nominal exposure concentrations (OECD 1992; USEPA 1996b; ASTM 2007). However, nominal concentrations represent a substantial portion of existing acute toxicity data. In a broad query of over 200,000 acute toxicity records for all aquatic taxa in the Ecotox database, 66% were based on nominal concentrations and 11% of records did not report if concentrations were measured or nominal. Nominal test data used in the present study comprised 71% of the acute toxicity records. To our knowledge, a comprehensive and quantitative comparison of toxicity data reported as either nominal or measured concentrations has not previously reported. We compared toxicity data reported as nominal and measured

concentrations for 227 species and chemical pairs, which had median nominal to measured ratios of less than 2.0 for all taxa. The slopes of log-linear models relating measured to nominal concentrations were not significantly different than 1.0 for all well-represented taxa except crustaceans and all taxa combined using all acute test types. However, the slope in the latter comparison was 0.97 with an R^2 of 0.96 and was significantly different from one because of its low standard error (0.014). ICE models were developed for 12 pairs of fish species using identical chemical composition and varying only on use of toxicity reported as measured or nominal concentrations. While model slopes and intercept were variable among measured and nominal models, they were not significantly different for any species pairs and models developed from one concentration type predicted the reported toxicity of the other with high certainty. These results indicate that acute toxicity test results reported as either measured or nominal concentrations may be included in ecotoxicological models.

Organism life stage is well recognized as an important determinant of intra-species variability in toxicity test results. Early life stages have been considered the most sensitive based on the analysis of chronic toxicity test results, and early life stage testing has been adopted as a surrogate for longer term chronic toxicity tests (Macek and Sleight 1977; McKim 1985; Hutchinson et al. 1998). This study also found generally greater sensitivity of early life stages as shown by median toxicity ratios greater than 1, calculated as the older life stage divided by the younger life stage. However, test guidance for acute toxicity tests typically recommend juvenile life stages in efforts to standardize methods and results for a greater number of test species (e.g., OECD 1992; USEPA 1996b; ASTM 2007). Comparisons of acute toxicity of different life stage have been conducted in many studies; however, most contain small sample sizes of a limited number of species and chemicals (Fisher et al. 1994; Gaikowski et al. 1996; Medina et al. 2002; Wang et al. 2007) or do not include juvenile and adult stages (Mayer and Ellersieck 1986). To our knowledge, a comprehensive and quantitative comparison of life stage sensitivity in acute toxicity tests has not previously reported. Our results indicate that while early life stages were consistently more sensitive than later life stages, median ratios were generally less than 2.0. The ranges of ratios showed a maximum difference in adult and larval fish, which differed by a factor of 45, however, the interquartile ranges of ratios showed much less variation across life stages in general. Well represented log-linear models ($N > 30$) all contained slopes near 1.0 and were not significantly different from 1.0 for all comparisons except that for juveniles and adults of all taxa combined. Overall, these results indicate that life stage contributes a significant amount of variation in untransformed data, but is likely to

be a minimal source of variation in log-transformed data. Additional data and analyses are needed to assess the relative sensitivity of the embryo stage of aquatic organisms.

Ecotoxicological models based on transformed data for which standardization may be a concern include ICE models, Species Sensitivity Distributions (SSDs) and acute to chronic time-response models. ICE models were used as an example ecotoxicological model in this study because their relatively small data requirements allowed paired models with identical chemical composition to be developed using only flow-through and static tests or toxicity reported as either measured or nominal. SSDs are cumulative probability distributions that maybe used to derive hazard levels for risk assessment and have large data requirements (Newman et al. 2000). Because of the large data needs of SSDs, most authors have not standardized the data included in their distributions for the factors analyzed in the present study (Versteeg et al. 1999; Newman et al. 2000; De Zwart 2002; Dyer et al. 2006, 2008) while others standardized for age class to limit intraspecific variability (Buckler et al. 2005; Raimondo et al. 2008). Acute to chronic time-response models use the time course of mortality measured in acute toxicity tests to estimate chronic toxicity and general standardization of acute test type has been applied in some studies (Mayer et al. 1994; Mayer et al. 2002). Datasets used in time-response modeling may require additional standardization, including considering the time course of mortality (Barron et al. 2008).

Our analyses of the influence of acute test type, life stage, and concentration reporting contained some inherent uncertainties. Some caution should be applied when interpreting results comparing slopes to 1.0 using fewer than 30 datapoints. Power analysis indicates that 33 datapoints are required for a one sample t -test to maintain sufficient power ($1-\beta = 0.8$), which minimizes the probability that significant differences are not detected. For example, the comparison of toxicity obtained from flow-through and static tests for insects had a slope of 0.85, which was not determined to be significantly different from one. In this example, variability around the slope (standard error of slope = 0.153) was too large and sample size ($N = 6$) was too small to detect a significant difference between the slope and 1.0. Conversely, the log-linear model for toxicity reported as measured and nominal for all taxa and all acute test types ($N = 227$) had a slope of 0.97 and was found to be significantly different from one due to low variance around the slope (standard error of slope = 0.014). These conflicting results indicate that the statistical analyses should be supplemented with a qualitative assessment of the modeled relationships. The ICE models developed using different test types and reported concentration types showed variable slopes and intercepts,

but no significant differences were found. Differences maybe undetected when sample sizes are small or variability is high. However, the models still predicted toxicity for other test or concentration type with relatively high certainty and serve to demonstrate acute test type or reported concentration are less critical factors by which to standardize data for ICE model development.

Our objectives focused on assessing the variability associated with log-transformed toxicity data used in model development and was not intended to guide decisions for species specific or chemical specific risk assessments. This research was intended to benefit future model development efforts by determining standardization needs and criteria for large datasets in an effort to streamline and maximize data acquisition efforts. Our results concluded that, because log-transformations reduce data variability, life stage, acute test type, and reported concentration type may not be a critical factors for standardization in ecotoxicological models that address large scale relationships. However, the range of toxicity ratios indicated that life stage can contribute a large amount of variability to untransformed toxicity data, particularly for fish, and standardization of life stage should be regarded on a species or taxa-specific basis when applying untransformed data in chemical or species specific risk assessments.

While standardizing data is desirable to reduce extraneous sources of variation so models reflect species sensitivity relationship with greater certainty (Buckler et al. 2003), it can reduce the amount of data available for model development. For example, Raimondo et al. (2007b) demonstrated that standardizing ICE models by chemical mode of action reduced model error in only some models, but decreased the overall number of robust models. Models developed from database compilations have the potential to capture chemical and species diversity, which may be lost if data standardization criteria are applied to factors that may not contribute a significant source of variation. While the factors analyzed in this study are likely a significant source of variation for untransformed toxicity values that maybe used in risk assessment, this variation is less important when data are log-transformed. While this study did not comprehensively evaluate all model applications, our results demonstrate that the variability in log-transformed acute toxicity data associated with acute test type, reported concentration type, or life stage is not likely to influence robustness of ecotoxicological models.

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